Mason Bane

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Summary: Computer Engineering student with hands-on hardware experience in circuit design, PCB prototyping, and microcontroller systems. Proficient in hardware analysis tools and low-level programming for complete hardware/software integration. Brings research background and analytical skills to hardware design challenges.

Education

Tarleton State University — Stephenville, TX
Bachelor of Science in Computer Science, Concentration in Computer Engineering — *Aug. 2022 – May 2026 (Expected)*
Minor in Mathematics
GPA: 3.94/4.00 (Institutional) — Cumulative: 3.75/4.00

Hill College — Hillsboro, TX
Associate of Arts in Liberal Arts — *Aug. 2019 – Sept. 2023*

Experience

Undergraduate Research Assistant - Lead Programmer — May 2024 – Present
Tarleton State University — *Stephenville, TX*

- Developed and optimized of various N-Body simulations utilizing C/C++, enhancing computational efficiency and accuracy through advanced algorithms and parallel processing with CUDA.
- Collaborated with interdisciplinary teams to create digital twins to model complex problems using OpenGL and Blender, translating scientific concepts into interactive simulations and improving data interpretation.
- Conducted rigorous testing and debugging of simulation software, ensuring robust performance and reliability while documenting processes to facilitate knowledge transfer and future research initiatives.

Undergraduate Technology Specialist - HPC Lab Manager — Aug. 2024 – Present
Tarleton State University — *Stephenville, TX*

- Managed and maintained 15 Linux devices in a high-performance computer lab dedicated to research initiatives
- Performed system updates, hardware maintenance, and technical support, resolving issues promptly to minimize downtime and maintain optimal performance.
- Maintained a clean and organized lab environment, promoting a collaborative workspace that fosters innovation and productivity.

Projects

Secure IoT Sensor Node with Hardware TrustZone | *C, STM32CubeIDE/HAL, I2C/SPI* — Dec. 2025 – Present

- Implementing a secure data pipeline on an STM32H5 MCU, utilizing Arm TrustZone and the Secure Manager to create a hardware-rooted chain of trust for environmental sensor data.
- Developing drivers and application logic to interface with a Bosch BME280 temperature, humidity, and pressure sensor via I2C communication protocol.
- Structuring firmware to cryptographically sign sensor readings within the secure processing environment for tamper-evident logging.
- Utilizing STM32CubeIDE and the HAL for peripheral configuration and hardware security module (HSM) management as a foundation for secure telemetry systems.

Analog Pink Noise Generator Circuit | *LTSpice, Python, MATLAB, Breadboarding* — Aug. 2025 – Dec. 2025

- Designed and prototyped a BJT-based pink noise generator with active filter stage using LTSpice for simulation and breadboarding for physical implementation.
- Developed Python scripts to quantitatively compare simulation results with an ideal pink noise spectrum, guiding component selection and filter tuning for improved accuracy.
- Built a physical demonstration circuit with audio output, successfully verifying simulation models against real-world performance.
- Collaborated in a two-person team to present comprehensive design methodology, results, and data-driven decision process to peers and professors.

Technical Skills

Hardware Platforms: STM32 (H5/Cortex-M33), TIVA-C (TM4C), Arduino, Raspberry Pi, ARM Cortex-M
Firmware Development: C, C++, ARM Assembly, I^2C, SPI, UART, STM32CubeIDE/HAL, RTOS
Circuit Design & Tools: LTSpice, Analog/Digital Circuit Design, PCB Prototyping, Oscilloscope, Multimeter
Development Tools: Git, Linux, CMake/Make, GDB, Python, Bash, VS Code
Simulation & Research: MATLAB, CUDA, OpenGL, Blender